

■ FinanceFlow — AI Finance Controller

Track 04: 'Run the books and the cash position' | Comprehensive Guide & Video Pitch

1. Executive Summary & The 2026 Core Bottleneck

In modern financial operations, **verification capacity, not generation speed, is the primary bottleneck**. While LLMs can draft text easily, closing the actual finance-ops loop across multi-source ledgers, resolving settlement discrepancies, and forecasting cash positions requires strict accuracy, deterministic reconciliation, and an honest exception handling engine. FinanceFlow was built specifically to solve this.

2. Track 04 Alignment Matrix

Recruiter Requirement (Track 04)	FinanceFlow Implementation	Status
Multi-source Reconciliation (50+ record synthetic batch)	Automated 3-way matching across Bank Statements, ERP Ledgers, and Payment Gateway records.	100% Ready
Measured Accuracy & Exception List ('One cherry-pick proves nothing')	Calculates real match rate (96%), throughput (tx/sec), and outputs detailed categorized exceptions (fee mismatches, unposted invoices).	100% Ready
Settlement Q&A; Agent	Conversational AI context engine answering real-time ledger settlement and discrepancy queries.	100% Ready
Forward Cash Forecaster	30/60/90-day cash runway and multi-year compound wealth simulation engine.	100% Ready
Tax-Line Matcher & KYC	Old vs New regime tax deduction classifier + regex PAN card validation.	100% Ready

3. ■ 5-Minute Video Demo Script (Minute-by-Minute)

■ Minute 1 [0:00 – 0:45] — Hook & Problem Statement

■ Visual Cue: Open Dashboard (<http://localhost:5500>) showing KPI Cards and GitHub Repo.

"Hello everyone, I'm Ritik Pandey, and this is my submission for Track 04: AI Finance Controller — Run the books and the cash position. In 2026, builder consensus is clear: generation speed is no longer the bottleneck—verification capacity is. Finance teams still spend countless hours manually reconciling multi-source ledgers, tracking settlement discrepancies, and forecasting forward cash positions. To solve this, I built FinanceFlow—an autonomous AI Finance Controller and Finance-Ops engine that automates multi-source reconciliation across 50+ record batches, reports measured accuracy with an honest exception list, provides a real-time Settlement Q&A; Agent, and forecasts dynamic cash positions."

■ Minute 2 [0:45 – 2:00] — Multi-Source Reconciliation & Honest Exception List

■ Visual Cue: Click 'Run 50+ Record Batch Auto-Reconciliation' and point out match rate and exceptions.

"Let's dive straight into the core track requirement: Multi-Source Reconciliation. Here, the engine ingests a synthetic batch of 50+ multi-source transactions across three distinct data pipelines: Bank Statements, ERP Invoices, and Payment Gateway logs. Notice the bar we set for this engine: First, it achieves a measured match rate of 96% with automated 3-way hash matching. Second, more importantly, it does not cherry-pick. It surfaces an Honest Exception List highlighting unresolved records—such as a \$150 gateway fee mismatch or an unposted ERP invoice. Each exception is categorized with an exact discrepancy reason so human controllers can audit or auto-resolve with one click."

■ Minute 3 [2:00 – 3:00] — Settlement Q&A; Agent & Forward Cash Forecaster

■ Visual Cue: Open AI Chat drawer, type settlement query, then show Cash Runway chart.

"Next is our Settlement Q&A; Agent. Connecting directly to our in-memory financial context service, the AI assistant answers real-time ledger questions such as where our cash bottlenecks are or why specific payouts are delayed. Moving to the Forward Cash Forecaster: instead of relying on static spreadsheets, our forecasting model projects 30, 60, and 90-day cash runway based on recurring debt obligations, verified income streams, and dynamic burn rates, giving leadership a clear visual of their future liquidity."

■ Minute 4 [3:00 – 4:15] — Full FinTech Suite & Architecture

■ Visual Cue: Show Investment Portfolio, Tax Estimator, and backend code in IDE.

"Beyond reconciliation, FinanceFlow is a comprehensive FinTech platform featuring Tax-Line Matching for automated deductions under Old vs New tax regimes, an Investment Portfolio Asset Tracker with multi-scenario win/loss recovery projections, and Sandbox PAN & Income Verification. Under the hood, the architecture is enterprise-grade: built on Java 17 and Spring Boot 3.2.1 with strict Spring Security 6 JWT authentication, token blacklisting, PostgreSQL/H2 persistence, and containerized Docker/Kubernetes manifests."

■ Minute 5 [4:15 – 5:00] — Conclusion & Wrap-Up

■ Visual Cue: Show README.md on GitHub, commit log, and active app.

"To summarize: FinanceFlow meets and exceeds Track 04's bar by closing the finance-ops loop with high throughput, measured verification accuracy, conversational settlement intelligence, and honest exception handling. The complete source code, docker-compose configuration, and documentation are available on my GitHub repository. Thank you for your time, and I look forward to discussing this in the next round!"

4. ■ Top Technical Q&A; (For Evaluators & Interviews)

Q1: How does your reconciliation engine handle timing and fee discrepancies?

A: It implements a multi-pass heuristic matcher: Pass 1 does strict 3-way hash matching (ID + exact amount). Pass 2 applies a configurable tolerance delta for gateway processing fees (e.g. 2% MDR) and settlement date windows (+/- 48 hours). Records failing both passes are explicitly classified into the Honest Exception List.

Q2: Why did you prioritize measured accuracy and exception classification over simple matching?

A: In real-world enterprise finance, an engine that blindly marks 100% matches creates audit hazards. An honest exception list with root-cause categorization (AMOUNT_MISMATCH, UNPOSTED_ERP, GATEWAY_TIMING) gives finance controllers full trust and auditability.

Q3: How is the backend secured and architected?

A: The backend runs Spring Boot 3.2.1 with Spring Security 6. All protected routes require a cryptographically signed HMAC-SHA256 JWT Bearer token with server-side token blacklisting on logout. Data access is managed via Spring Data

JPA and Hibernate ORM on PostgreSQL/H2.

Q4: What is the purpose of the Settlement Q&A; Agent?

A: It bridges the gap between raw ledger records and human decision-makers by aggregating real-time ledger states into a conversational context window to answer ad-hoc questions on cash positions and pending settlements.